

Evidence Summary - Technology #07

Self-Powered Sensing via Piezoelectric Energy Harvesting

External Evidence Sources (5 total):

Source 1: Conceptual Piezoelectric-Based Energy Harvester from In

URL: <https://pmc.ncbi.nlm.nih.gov/articles/PMC11434573>

Excerpt: by M Khazaei · 2024 · Cited by 11 — This paper studies the development of piezoelectric energy harvesting for self-powered leadless intracardiac pacemakers .

Source 2: Piezoelectric Energy Harvesting for Pacemaker Applications

URL: <https://pubs.acs.org/abseba/article/11/8/4570/3599553/Piezoelectric-Energy-Harvesting-for-Pacemaker>

Excerpt: by D Singh · 2025 · Cited by 13 — This device design produced a mean power of 0.78 μ W for a heart rate of 84 bpm when implanted in the pig's heart at the right apical ventricle.

Source 3: Recent advancements in piezoelectric energy harvesting

URL: <https://journals.sagepub.com/doi/10.1177/1045389X231200144?int.sj-full-text.similar-articles.3>

Excerpt: Oct 21, 2023 — This work focuses on EH for biomedical devices such as pacemakers, gastric stimulators, implantable cardioverter-defibrillator (ICD), etc. A ... Read more

Source 4: Implant harvests heartbeat power | Research

URL: <https://www.chemistryworld.com/news/implant-harvests-heartbeat-power/7002.article>

Excerpt: Jan 21, 2014 — A flexible piezoelectric implant that harnesses energy from the body's natural motions has been developed by researchers in the US and China. Read more

Source 5: Cardiac energy harvesting and sensing based on

URL: <https://www.sciencedirect.com/science/article/am/pii/S2211285520306534>

Excerpt: by L Dong · 2020 · Cited by 127 — The essential energy harvesting mechanism for those reported devices is to harness cardiac motion or pacemaker lead motion into strain within the piezoelectric ...

*Evidence Classification: This dossier contains source-native information, derived engineering analysis, engineering proposals, and unresolved questions.
No evidence has been physically validated.*